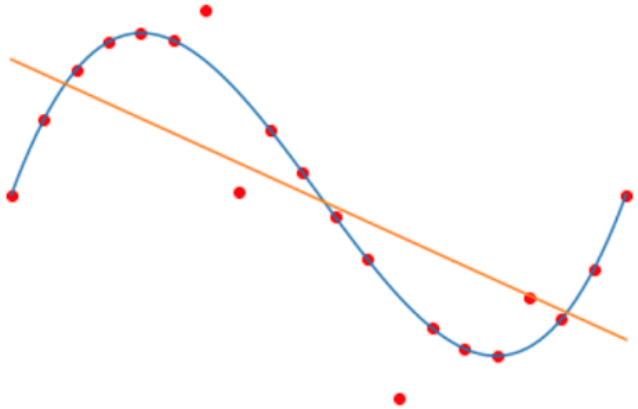
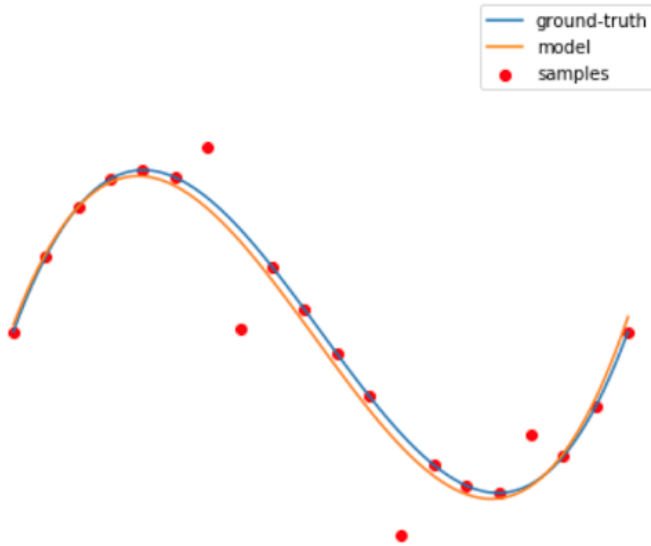
Why? How?

Case 1: Bias-Variance trade-off and Double Descent

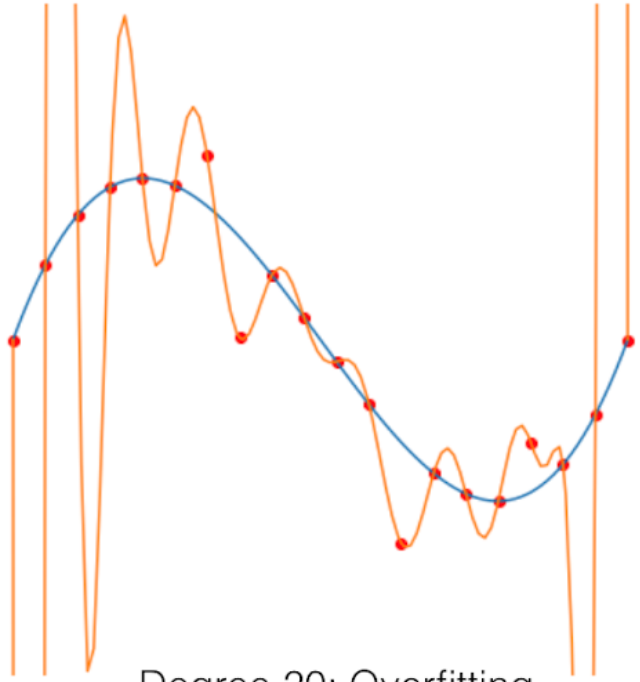
Why?



Degree 1: Underfitting



Degree 3: Right fit

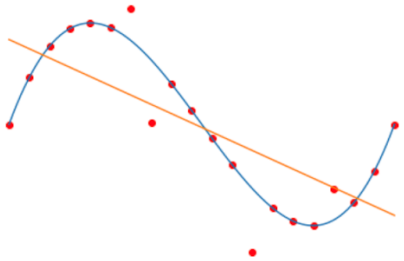


Degree 20: Overfitting

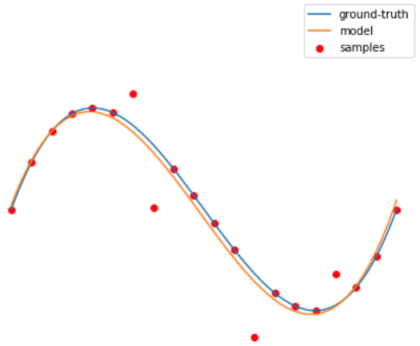
[source: https://openai.com/blog/deep-double-descent/]

Case 1: Bias-Variance trade-off and Double Descent

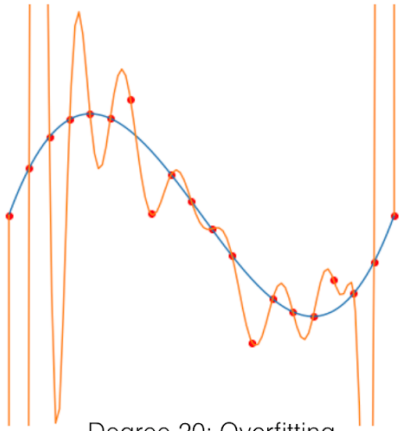
Why?



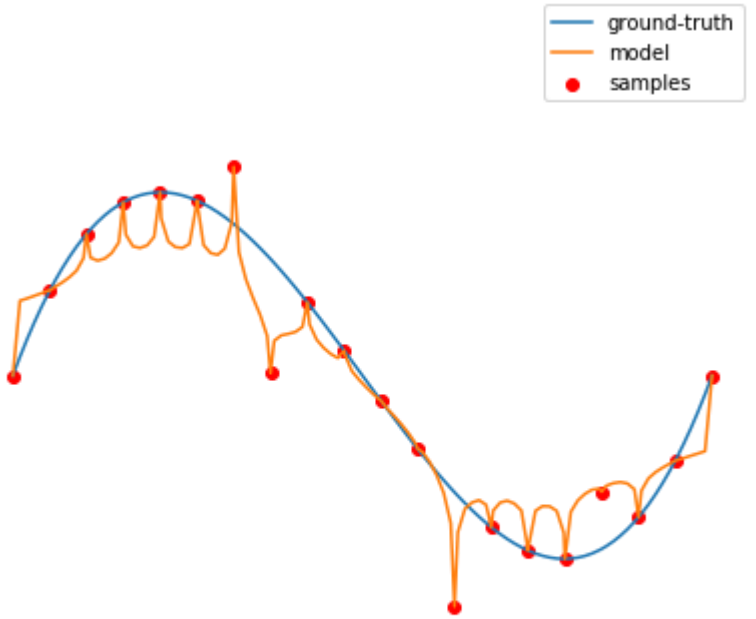
Degree 1: Underfitting



Degree 3: Right fit



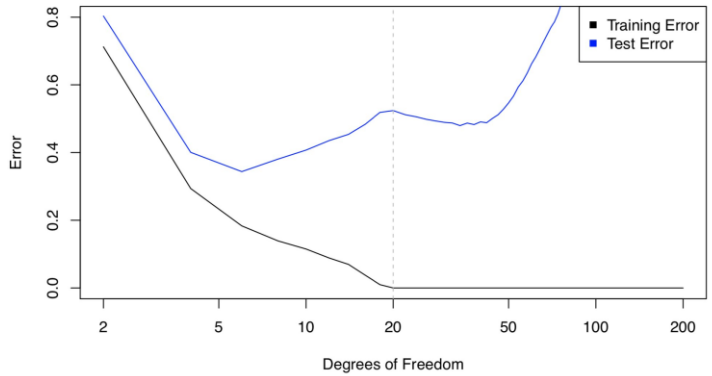
Degree 20: Overfitting



Degree 1000: overfitting?!

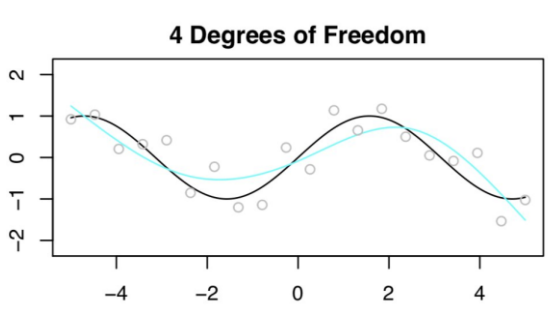
[source: https://openai.com/blog/deep-double-descent/]

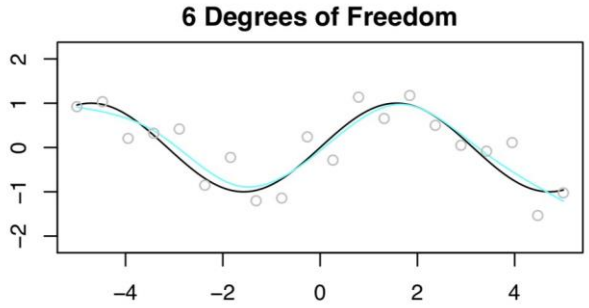
Case 1: Bias-Variance trade-off and Double Descent

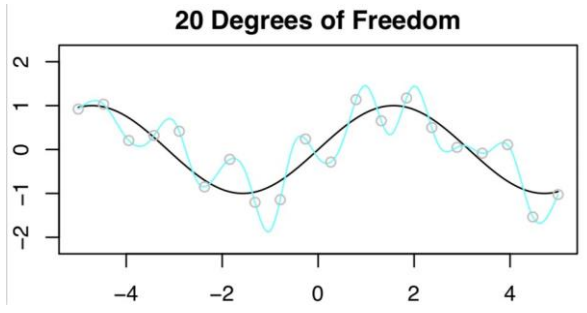


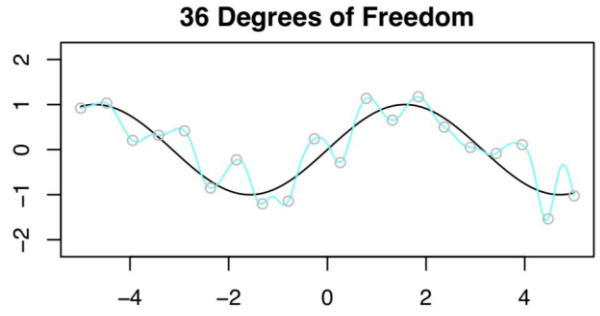
Why?

- 20 datapoints (generated from sin(x) + some noise
- Approximating functions using Splines with d basis functions (degrees of freedom)
- With 20 basis functions there is exactly 1 solution to the problem
- From point 20 on, we have many perfect solutions, so we use some regularization, *e.g.*, minimum norm fit









[source: Twitter thread by Daniela Witten: https://twitter.com/daniela_witten/status/1292293102103748609]

Case 1: Bias-Variance trade-off and Double Descent

Open questions

- what are exactly the regularization which is applied in neural networks?
- Is dropout and/or weight decay really necessary for training deep nets?
- How can we understand double descent in light of the fact that “we always have the regularization on for neural networks”.
- what is the effective degrees of freedom for a neural network? considering all aspects such as architecture (symmetry, …), data (low-dimensional manifold, …), and optimization
- …

This gets us to the 2nd intriguing case